

James Hobin

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github.com/hobinjk hobinjk.github.io/resume

EDUCATION	Massachusetts Institute of Technology 2012–2017 - Obtained Master of Engineering in Computer Science from the MIT Media Lab in 2017 - Obtained Bachelor of Science in Computer Science in 2016
EXPERIENCE	PTC Inc. distributed systems engineering augmented reality industrial iot artificial intelligence 2019–2024 <i>Senior Principal Innovation Engineer</i> - Created a machine-learning-enabled tool which cuts the time for factory work analysis by 80% - Developed a container-based cloud host for real-time multi-user contextual XR experiences - Invented depth streaming protocol with peer-to-peer connections and storage format - Led testing and developer tooling efforts across all Reality Lab projects - Granted two patents for original research (US11776205B2, US11625140B2) 2nd author on CHI 2024 publication "The Pop-Up Metaverse: A Multi-User, Multi-Tasking Spatial Computing Environment for Collaborative Spatial Problem-Solving" Mozilla javascript node linux embedded devices c 2017–2019 <i>Platform Engineer, Advanced Development</i> - Developed a C library for any Arduino-compatible device to expose the Web of Things API - Created a comprehensive rules engine for automating smart home functionality - Designed, implemented, and tested the Mozilla IoT Gateway's OAuth security framework - Wrote a series of technical blog posts for Mozilla Hacks (90,000+ followers) MIT Media Lab c++ ios javascript node 2012–2017 <i>Fluid Interfaces Researcher</i> - Designed adaptable, modular interfaces for the Reality Editor, a novel AR IoT platform - Refined inter-device protocols for reliability and speed 2nd author on UBICOMP 2013 publication "Reality Editor: Programming Smarter Objects" Mozilla c++ javascript android internals 2014, 2015 <i>Firefox OS Systems Front End Team Intern</i> - Produced several major features and improvements over two summer internships - Spearheaded research project to uncover a significant market opportunity for Firefox OS
PROJECTS	New XKit javascript browser extension open source 2015–2019 - Forked and revitalized a popular open source project for improving Tumblr - Supported over 400,000 users across Chrome, Firefox, and Safari Art and Games javascript c++ rust writing webgl 2010–Present - League of Lines: Voice-line-based online competitive card game - Full portfolio: https://hobinjk.github.io/portfolio Cabbage Legs javascript machine learning canvas 2015 - Programmed a genetic algorithm that optimizes simulated quadrupedal robots - Extended into an online game for crowdsourcing the design of robots as undergraduate researcher at CSAIL's Distributed Robotics Laboratory